THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of 1s in the binary form of 150 is: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

2. The Roman numeral for 2449 is: 1

(a) MMCDXLIX **(b)** MMCDXLVIII **(c)** MMCDL **(d)** MMCDLIX

3. The number that comes immediately after 11011_2 in the binary system is: 1

(a) 11100_2 **(b)** 11010_2 **(c)** 11101_2 **(d)** 110110_2

4. A number written in base 5 ending in the digit 0 is always divisible by: 1

(a) 10 **(b)** 5 **(c)** 6 **(d)** 4

5. If $(43)_b = 31$ in decimal (a two-digit number in base b), then b equals: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

6. Weights of 1, 2, 4, 8, ... kg (powers of 2) are available. The least number of weights needed to weigh any whole number of kg up to 63 kg is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 8

7. Which of the following numbers is the greatest? 1

(a) 76_8 **(b)** 203_5 **(c)** 60_{10} **(d)** 111101_2

8. The number of 1s in the binary form of 255 is: 1

(a) 9 **(b)** 8 **(c)** 10 **(d)** 7

9. **Assertion (A):** The Babylonian number system had base 60. 1

Reason (R): The Maya number system also used base 10.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The Roman numeral IC represents 99. 1
Reason (R): In Roman numerals, I can be subtracted only from V and X.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write 157 in binary and count the number of 1s in it. 2

12. Convert the Roman numeral CCXLIV to binary. 2

13. Find the base b if $(46)_b = 38$ in decimal. 2

14. Write 203 in binary and count the number of 1s in it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 102 g? Explain using binary numbers. 3

16. Add 1111_3 and 1002_3 (numbers in base 3) and give the answer in base 3. 3

17. How many different numbers can be written using at most 4 digits in base 5? What is the largest of them in decimal? 3

OR

A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 89 g? Explain using binary numbers. 3

18. Add 335_6 and 221_6 (numbers in base 6) and give the answer in base 6. 3

OR

How many different numbers can be written using at most 5 digits in base 4? What is the largest of them in decimal? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Gopal runs a grain shop in Lucknow and owns weights of 1, 2, 4, 8 and 16 kg, each only once. He wants to weigh 13 kg and 27 kg of grain using binary numbers. 5
(i) Write 13 in binary.
(ii) Which weights are used for 13 kg?
(iii) Write 27 in binary.
(iv) Show that these five weights can measure every whole number of kg from 1 to 31.

20. In some base b the number $(257)_b$ equals 175 in decimal. Find b and verify your answer by expanding. 5

OR

Subtract $2212_3 - 1022_3$ in base 3. Verify your answer in decimal. 5

END OF QUESTION PAPER — Total: 40 Marks